AQ3.2: Activity Questions 2 - Not Graded

This assignment will not be graded and is only for practice.
Level 1

If $(1, 2, 3) + v = (3, 2, 1)$ in the vector space R^3 with usual addition and scalar multiplication, where $v = (a, b, c)$, then find the value of $a + b + c$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 0
1 point
Which of the following statements are related to the cancellation law for vectors x, y, z in a vector space.

Statement P: $x + y = z + y \implies x = z$.
Statement Q: $x + y = z + y \implies x = y$.
Statement R: $x + y = z + y \implies y = z$.
Statement S: $x + y + z = x \implies y + z = 0$

Find the number of correct statements

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 2
1 point
1 point
Which of the following options is/are true for a non zero vector space V ?

- [ ] $x + y = z + y \implies x = z \ \forall x, y, z \in V$.
- [ ] $ax = bx, \ \forall \ x \in V \implies a = b,$ where $a, b \in R$.
- [ ] $ax = ay, \ \forall \ a \in R \implies x = y,$ where $x, y \in V$.
- [ ] None of the above.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$x + y = z + y \implies x = z \ \forall x, y, z \in V$.
$ax = bx, \ \forall \ x \in V \implies a = b,$ where $a, b \in R$.
$ax = ay, \ \forall \ a \in R \implies x = y,$ where $x, y \in V$.
1 point
Which of the following options is/are true for a non zero vector space V ?

- [ ] $av = 0, \ \forall \ v \in V \implies a = 0$.
- [ ]

$av = 0, \forall\, a \in R \implies v = 0$av = 0, ∀ a ∈ R ⟹ v = 0.

☐

$c \in R$c ∈ R and $v, w \in V$v, w ∈ V such that, $c(v - w) \neq cv - cw$c(v − w) = cv − cw.

☐

$c((v + w) - (v - w) - 2w) = 0\ \forall c \in R,$ and $v, w \in V$c((v + w) − (v − w) − 2w) = 0 ∀c ∈ R, and v, w ∈ V

No, the answer is incorrect.

Score: 0

Accepted Answers:

$av = 0, \forall\, v \in V \implies a = 0$av = 0, ∀ v ∈ V ⟹ a = 0.

$av = 0, \forall\, a \in R \implies v = 0$av = 0, ∀ a ∈ R ⟹ v = 0.

$c((v + w) - (v - w) - 2w) = 0\ \forall c \in R,$ and $v, w \in V$c((v + w) − (v − w) − 2w) = 0 ∀c ∈ R, and v, w ∈ V

Let $(3, 2, -3) \in R^3$(3, 2, −3) ∈ R3 be a vector from the vector space R^3R3 with usual addition and scalar multiplication. If $c(3, 2, -3) = (-6, -4, 6)$c(3, 2, −3) = (−6, −4, 6), find the value of cc.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) -2

1 point

1 point

Which of the following options is/are true?

☐ The zero vector is a unique vector in a vector space.

☐ Inverse of a vector is a unique vector in a vector space.

☐

$0v \neq 00$v = 0, where vv is a vector from a vector space of RR.

☐

$c0 = 0$c0 = 0, where $c \in R$c ∈ R and 0 is the zero vector from a vector space over RR.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The zero vector is a unique vector in a vector space.

Inverse of a vector is a unique vector in a vector space.

$c0 = 0$c0 = 0, where $c \in R$c ∈ R and 0 is the zero vector from a vector space over RR.

Level 2

Let VV be a plane plane parallel to the XYXY -plane. Any plane parallel to XYXY -plane is given by $z = c$z = c. We define addition of two vectors $v_1 = (x_1, y_1, c)$v1 = (x1, y1, c) and $v_2 = (x_2, y_2, c)$v2 = (x2, y2, c) on VV as follows: First project v_1v1 and v_2v2 on the XYXY -plane (we will get the vectors $(x_1, y_1, 0)$(x1, y1, 0) and $(x_2, y_2, 0)$(x2, y2, 0) by projection on the XYXY -plane) and then calculate the addition of the vectors we obtained by the projection on the XYXY -plane (we will obtain $(x_1 + x_2, y_1 + y_2, 0)$(x1 + x2, y1 + y2, 0)). Then project the obtained vector back to the plane VV (we will obtain the vector $(x_1 + x_2, y_1 + y_2, c)$(x1 + x2, y1 + y2, c)).

Let VV be a plane which is parallel to the XYXY -plane, defined by $z = 2$z = 2. Let $v_1 = (1, 2, 2)$v1

$= (1, 2, 2)$ and $v_2 = (0, 3, 2)$ be in V. Answer questions 7 and 8.

If $v_1^{'} = (a_1, b_1, c_1)$ and $v_2^{'} = (a_2, b_2, c_2)$ are the projections of v_1 and v_2 on the XY - plane, respectively, then find the value of
$2(a_1 + b_1 + c_1) - (a_2 + b_2 + c_2)$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 3
1 point
If $v_1 + v_2 = (a_1, b_1, c_1)$ as per the addition defined above, then find the value of $(2a_1 + b_1 - 2c_1)$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 3
1 point

Let V be a plane plane parallel to the XY -plane. Any plane parallel to the XY -plane is given by $z = c$. We define scalar multiplication of a vector $v = (x, y, c)$ on V as follows: First project v on the XY -plane first (we will get the vector $(x, y, 0)$ by projection on the XY -plane) and then calculate the scalar (α) multiple of the vector we obtained by the projection on the XY -plane (we will obtain $(\alpha x, \alpha y, 0)$). Then project the obtained vector back to the plane V (we will obtain the vector $(\alpha x, \alpha y, c)$).
Let V be a plane which is parallel to the XY -plane, defined by $z = 2$. Let $v = (1, 2, 2)$ be in V. Answer questions 9 and 10.

If $v^{'} = (a_1, b_1, c_1)$ is the projection of v on the XY - plane, then find the value of $(a_1 + 2b_1 + c_1)$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 5
1 point
If $4v = (a_1, b_1, c_1)$ as per the scalar multiplication defined above, then find the value of $(a_1 + 2b_1 + c_1)$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 22
1 point

Check Answers
Your score is: 0/10